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Author(s): Donald A. Dewsbury and Alan W. Hodges

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COPULATORY BEHAVIOR AND RELATED PHENOMENA IN SPINY MICE (*ACOMYS CAHIRINUS*) AND HOPPING MICE (*NOTOMYS ALEXIS*)

DONALD A. DEWSBURY AND ALAN W. HODGES

Department of Psychology, University of Florida, Gainesville, FL 32611

ABSTRACT.—Spiny mice, *Acomys cahirinus*, are unusual muroid rodents in that females produce spontaneously active corpora lutea during estrous cycles. Their copulatory behavior is characterized by an incipient lock, no intravaginal thrusting, multiple intromissions preceding ejaculation, and multiple ejaculations, but with a low number of ejaculations per test. Hopping mice, *Notomys alexis*, are unusual with respect to the small testis size, reduced complement of accessory glands, and large penile spines, all of which combine to predict a locking pattern of copulation. As predicted, hopping mice display a true lock. They also display a lack of both intravaginal thrusting and multiple intromissions, and generally cease copulating with the occurrence of the first lock. These and supplementary data can be combined to generate a prediction of monogamy in the field for hopping mice and a similar, but much weaker, tendency for spiny mice.

Species-typical patterns of copulatory behavior typically are correlated and integrated with variation in reproductive physiology and anatomy (e.g., Dewsbury, 1975). In nearly all species of muroid rodents for which characteristics of the female estrous cycle are known, either ovulation or the occurrence of a functional luteal phase, or both are contingent on copulatory activity or similar vagino-cervical stimulation. A major function of male copulatory behavior lies in triggering female neuroendocrine reflexes underlying ovulation or a functional luteal phase; male copulatory behavior and female stimulus requirements appear to have evolved in a coordinated manner (Dewsbury, 1978). It has been proposed that in species in which functional corpora lutea form spontaneously, male copulatory behavior may be less elaborate, as there would be little function to vagino-cervical stimulation (Bignami and Beach, 1968). The two murid species for which such active luteal phases have been described are the Australian species, *Mesembriomys gouldii* (Crichton, 1969), and spiny mice, *Acomys cahirinus* (Peitz, 1981). Spiny mice are reported to have an average cycle length of 11.1 days with a uterus that spontaneously supports the decidual response to uterine trauma.

The estrous cycles of hopping mice, *Notomys alexis*, by contrast, are typical of the subfamily Hydromyinae, with mean lengths of approximately 7–9 days (Breed, 1975, 1976). Male reproductive anatomy is remarkable (e.g., Breed, 1981a) because although there is a large ventral prostate gland, the seminal vesicles, coagulating, preputial, and dorsal prostate glands are small and vestigial and the testes also are small (Breed, 1980, 1981a, 1981b, 1982). The glans penis is small, with very large penile spines (Morrissey and Breed, 1982). There appears to be either no, or a minute vaginal plug formed at copulation (Breed and Sarafis, 1979; Breed, 1980, 1985). In muroid rodents with “simple” glands (Hooper and Musser, 1964), species with a copulatory pattern including “locks” (i.e., a mechanical tie between penis and vagina) generally have thick glands, elaborate penile spines, reduced complements of accessory glands, and a lack of copulatory plugs (Dewsbury, 1974a, 1975; Hartung and Dewsbury, 1978). To the extent that a relationship found among great apes can be generalized, one might expect species in which males have relatively small testes to have reduced copulatory behavior and internal sperm competition (Short, 1979). These various anatomical data can be combined to generate a behavioral prediction that the copulatory pattern of *Notomys alexis* should be one with a lock and relatively little copulatory activity. Although locking patterns have been described in various species of Cricetidae, there appear to be no reports of locking in Muridae. Although Happold (1976) observed copulation in hopping mice, she did not report the occurrence of a lock. She did, however, report that copulatory activity ceases after one copulation.

The first quantitative studies of copulation in spiny mice and hopping mice are presented herein. In addition, various other information regarding dimorphism, development, and related

phenomena are presented in order that these species might be better understood in relation to social context and to other species (see Dewsbury, 1981a).

METHODS AND MATERIALS

Animals

Copulatory behavior was studied in 82 male and 80 female spiny mice (*Acomys cahirinus*) and 40 male and 47 female spinifex hopping mice, (*Notomys alexis*). Additional animals from successive litters in the breeding colonies were used in supplementary observations on body mass and age at eye opening. Spiny mice were derived from the colony maintained at the George Peabody College, Nashville, Tennessee. Hopping mice were derived from animals trapped in the southern Northern Territory, Australia by W. G. Breed. All animals that served as subjects were born in the laboratory and were adult (at least 90 days of age) at the time of testing. Animals were maintained in a windowless, air-conditioned room with a reversed photoperiod with 16L:8D of fluorescent light. They were housed in polycarbonate cages measuring 48 by 27 by 13 cm or 29 by 19 by 13 cm. Wood shavings were provided as bedding and water and Purina laboratory animal chow were provided ad lib.

Procedure

Tests of copulatory behavior were conducted in the home cage of the male (48 by 27 by 13 cm) and data were recorded on an Esterline-Angus operations recorder.

The copulatory behavior of both species was studied in both naturally-occurring and hormone-induced estrus. Estrous cycles were followed with daily vaginal smears taken with a thin wire loop. Tests were conducted on days on which there was a virtual absence of leukocytes in the smears; cornified and/or nucleated cells predominated. Tests in hormone-induced estrus were conducted following intramuscular injections of 0.06 mg estradiol benzoate 72 h before testing and 0.6 mg progesterone 6 h before testing.

Two procedures were used. In one males and females were housed in separate cages and estrous females were introduced into the male's home cage to initiate the test. When both this procedure and tests in the females' home cages yielded a low percentage of tests with copulation, males and females were housed across from each other on opposite sides of a wire mesh barrier; the barrier was removed to begin the tests. Most of the data were collected with the latter procedure. As there were no appreciable differences in the parameters of copulatory behavior using the two procedures, the data were combined. Overall, copulation was observed in spiny mice in 33 of 259 tests in natural estrus and 34 of 472 tests in hormone-induced estrus and in hopping mice in 15 of 87 tests in natural estrus and 39 of 215 tests in hormone-induced estrus.

Once tests were initiated animals were permitted a minimum of 1 h to begin copulating. If copulation was initiated, tests were continued to a criterion of at least 30 or 60 min without any copulations.

Copulatory behavior in spiny mice is typical of many rodent species (e.g., deer mice, *Peromyscus maniculatus*) in that it includes mounts (with no vaginal insertion), intromissions (with brief vaginal insertion but no sperm transfer), and ejaculations (with insertion and sperm transfer). These events are organized into discrete series with each series terminated by an ejaculation. The standard measures used with other species (e.g., Dewsbury, 1979) were used here and included: mount latency (ML)—time (sec) from the beginning of a test until the first mount or intromission; intromission latency (IL)—time (sec) from the beginning of a test until the first intromission; ejaculation latency (EL)—time from the first intromission of a series until ejaculation; intromission frequency (IF)—number of intromissions in a series; mount frequency (MF)—number of mounts in a series; mean inter-intromission interval (MII)—mean interval (sec) separating the intromissions of a series; postejaculatory interval (PEI)—time (sec) from ejaculation until the next intromission; and ejaculation frequency (EF)—the number of ejaculations in a test. Where more than one series occurred a number appearing as a suffix to a measure identifies the series to which the measure refers.

Because hopping mice have a locking pattern of copulation different measures were required. Only ML, IL, and EF were applicable. Added measurements included: lock duration—time (sec) in genital lock; post-lock time—time (sec) animals were watched after a lock; pre-lock mount frequency—number of mounts occurring before a lock; and post-lock mount frequency—number of mounts occurring after the lock.

RESULTS

Copulatory Behavior in Acomys cahirinus

Pre-copulatory interactions are typical of muroid rodents and culminate in the male following the female about the cage and eventually mounting her (see Dewsbury, 1975). When mounted,

TABLE 1.—*Parameters of copulatory behavior in spiny mice (Acomys cahirinus).*

Measure	Natural estrus mean ($\pm$ SE)	Hormone-induced estrus mean ($\pm$ SE)
Number of pairs	11	16
Number of tests	33	34
Mount latency	299.1 (53.8)	565.7 (98.9)
Intromission latency	538.9 (92.4)	790.2 (212.3)
Ejaculation latency—1	876.5 (79.3)	1,366.4 (215.6)
Intromission frequency—1	13.7 (1.2)	14.7 (2.4)
Mount frequency—1	3.6 (0.6)	3.6 (0.7)
Mean interintromission int.—1	71.7 (9.8)	92.7 (13.6)
Postejaculatory interval—1	1,542.8 (193.7)	2,133.3 (277.4)
Ejaculation latency—2	970.8 (222.8)	1,766.4 (523.4)
Intromission frequency—2	16.1 (7.2)	13.4 (2.8)
Mount frequency—2	2.8 (0.8)	7.7 (3.6)
Mean interintromission int.—2	97.4 (39.1)	225.6 (88.4)
Ejaculation frequency	1.3 (0.1)	1.2 (0.1)

the female displays a marked lordosis that appears necessary for penile insertion. The copulatory pattern is characterized by distinctive series of intromissions in which males mount the female, attain brief (<1 sec) vaginal insertion and display a ballistic dismount followed by genital grooming. Multiple intromissions appear prerequisite for ejaculation. The ejaculations are easily distinguished from intromissions and are remarkable among muroid rodents. There is an incipient lock, as the male typically dismounts moving back-to-back with the female with evidence of some resistance to separation as is characteristic of locking species. However, whereas in most species with true locks the lock duration is substantial and variable, in spiny mice it is generally brief and less variable, rarely lasting more than 2 sec. There is no repetitive pelvic thrusting during either intromission or ejaculation; the single thrust of insertion terminates thrusting activity. Although multiple ejaculations sometimes occur, an episode typically terminates with a single ejaculation.

Quantitative data on copulatory behavior in *Acomys cahirinus* in both natural and hormone-induced estrus are presented in Table 1. A set of *t*-tests revealed no significant differences as a function of mode of estrus. Initiation of copulatory activity was moderately rapid, with intromission latencies generally under 15 min. The first ejaculation was typically preceded by about 10–15 intromissions separated 1–2 min from each other. In natural estrus complete second series occurred in 11 of the 33 tests; copulation was resumed in 14 tests with PEIs under the standard 30 min criterion in 12 of those. In hormone-induced estrus there were complete second series in 8 of 34 tests; copulation was resumed in 14 tests, 3 of which had PEIs of 30 min or less. When they occurred, the quantitative characteristics of second series were not markedly different from those of first series.

Litters were delivered after 26 of the 33 tests in natural estrus. The mean gestation period was 38.1 days with a mean litter size of 2.0. A set of *t*-tests comparing parameters of copulatory behavior in the 26 fertile tests with the 7 not yielding a litter revealed no significant differences.

Copulatory Behavior in Notomys alexis

Pre-copulatory activity in hopping mice was as described by Stanley (1971) and Happold (1976) and was unremarkable for muroid rodents. The pattern of male chasing was more rapid than in spiny mice and insertion was generally preceded by a substantial number of mounts without insertion. At insertion the male gained a firm clasp on the female and the pair typically rolled on their sides in this posture. Sometimes the pair would struggle to separate, revealing the existence of a copulatory lock more substantial than that in spiny mice. The occurrence of the first lock terminated copulatory activity in 52 of the 54 tests in both kinds of estrus; a second lock was observed in the remaining two tests. Pelvic thrusting ceased at insertion and there were no intromissions preceding the lock/ejaculation.

TABLE 2.—Parameters of copulatory behavior in hopping mice (*Notomys alexis*).

Measure	Natural estrus mean ($\pm$ SE)	Hormone-induced estrus mean ($\pm$ SE)
Number of pairs	7	13
Number of tests	15	39
Mount latency	403.1 (223.9)	296.3 (86.0)
Intromission latency	943.4 (282.7)	995.7 (176.9)
Lock duration	16.7 (2.6)	33.8 (6.5)
Post-lock time	2,560.9 (115.8)	2,678.3 (183.3)
Pre-lock mount frequency	16.3 (4.9)	13.4 (2.8)
Post-lock mount frequency	2.5 (1.0)	1.6 (0.5)
Ejaculation frequency	1.0 (0.0)	1.05 (0.04)

Quantitative data on copulatory behavior in both natural and hormone-induced estrus are presented in Table 2. A set of *t*-tests revealed no significant differences between behavior in natural and hormone-induced estrus, although the difference in lock duration approaches significance ($P < 0.1$). The latency to the first lock was generally less than 10 min. Mean lock duration was 17 sec in natural estrus and 34 sec in hormone-induced estrus. Although pairs were watched for an average of over 40 min after the first lock, second locks were observed in just two tests in hormone-induced estrus. The second locks occurred in two different pairs and followed the first by 340 and 163 sec, respectively. Lock durations for these two locks were 44 sec and 163 sec; the latter was the longest lock observed in the study. The occurrence of a lock greatly reduced male pursuit and mounting of the female. In natural estrus, for example, there were means of 16 mounts in the 403 sec prior to locking but just 2.5 mounts of the 2,561 sec after the lock.

Litters were delivered after 8 of the 15 tests in natural estrus. There was a mean gestation period of 32.6 days, with a mean litter size of 4.4. A set of *t*-tests for the copulatory behavior in tests that were or were not followed by pregnancy revealed just one significant difference; females becoming pregnant received fewer pre-lock mounts than those not becoming pregnant (means 6.0 vs. 25.0, $t = 2.74$, *d.f.* = 13, $P \leq 0.02$).

Supplementary Data

Data on sexual dimorphism for body mass, analyzed using the methods of Dewsbury et al. (1980) are presented in Table 3. Spiny mice showed statistically significant sexual dimorphism, with males the heavier sex, at ages of both 35 and 90 days. No data are presented for hopping mice at 35 days because of difficulties in reliable sexing of individuals at that age. At 90 days of age females generally weighed somewhat more than males; but there was no significant difference between the sexes.

The age of eye opening was determined in *Notomys alexis* for 34 pups from 9 litters of 8 different adult pairs. The mean age at eye opening was 19.6 days (range: 18–21). The eyes of young spiny mice were consistently open on Day 1.

TABLE 3.—Sexual dimorphism for body mass in *Acomys cahirinus* and *Notomys alexis*.

Measure	<i>Acomys cahirinus</i>		<i>Notomys alexis</i>
	Day 35	Day 90	Day 90
Number of litters	25	25	18
Number of males	39	41	32
Number of females	32	32	35
Mean male body mass (g)	37.5	43.4	29.1
Mean female body mass (g)	33.7	37.8	29.8
<i>t</i>	4.57	7.45	1.28
<i>P</i>	0.001	0.001	n.s.
Cube-root ratio	0.96	0.96	1.01

TABLE 4.—Scores for *Acomys cahirinus* and *Notomys alexis* on the Monogamy Scale of Dewsbury (1981a).

Characteristic	<i>Acomys cahirinus</i>		<i>Notomys alexis</i>	
	Score	Scale value	Score	Scale value
1. Male dimorphism	Yes	−2	No	+2
2. Intromission latency	790.2	+1	296.2	−1
3. Allogrooming	—	0	—	0
4. Ejaculation frequency	1.2	+2	1.05	+2
5. Coolidge effect	—	0	—	0
6. Plug	Yes	−1	No	+1
7. Litter size	2.0*	+2	2.87*	0
8. Paternal behavior	Yes*	+1	Yes*	+1
9. Eyes open	1	−2	19.6	+2
Sum		+1		+7

* Data from Happold (1976), Makin and Porter (1984), Morrison et al. (1976), and Stanley (1971).

The reproductive tracts were examined using the methods of Baumgardner et al. (1982) on seven female hopping mice that had just copulated (21–75 min previously) in hormone-induced estrus. No plugs were found. Uterine horns were V-shaped and the interior walls of the cervix were deeply grooved. Mean lengths were: left horn 18.9 mm; right horn 20.2 mm; vagina 7.1 mm; cervix 5.8 mm (cf. Baumgardner et al., 1982). Two female spiny mice were sacrificed after they had copulated to satiety. The tracts of both females contained copulatory plugs that were lodged firmly. They measured 10 and 11 mm in length and 3.5 and 3.8 mm at maximum diameter.

These data, together with those from the literature, permit an analysis of these two species in relation to the “monogamy scale” proposed by Dewsbury (1981a) in an attempt to correlate laboratory data with social organization in the field. The data are presented in Table 4. The overall scores were +7 for *Notomys alexis* and +1 for *Acomys cahirinus* (cf. Dewsbury, 1981a).

DISCUSSION

Copulatory Behavior in *Acomys cahirinus*

The copulatory pattern of *Acomys cahirinus* is characterized by a lack of intravaginal thrusting and multiple intromissions prerequisite to ejaculation. Multiple ejaculations are delivered in some episodes. Spiny mice are difficult to classify with respect to locking, as there is a clear resistance to separation as in species with true locks, but the duration and extent of the lock were much less substantial than in other species. It would appear best to classify them as a locking species with the proviso that the locks are less developed than in other species. We would thus classify spiny mice as displaying pattern number 5 (lock, no thrusting, multiple intromissions, multiple ejaculations) in the schema of Dewsbury (1972). This is an unusual pattern; indeed, spiny mice are the only species in which we have found it. They are the only one of the 45 species studied with both a lock and a multiple intromission pattern. The stereotypy of the pattern is reflected, in part, in the invariance displayed as a function of mode of female estrus induction (natural vs. hormone-induced). This is typical of other species as well.

The reproductive tract of male spiny mice contains a normal, extensive complement of accessory glands (Peitz et al., 1979) and differs in this respect from species with true, functional locks (see Dewsbury, 1975). Testis size appears slightly below average for muroids but is not remarkable (Dieterich et al., 1973). It is not so small as to suggest the rather low number of ejaculations displayed (cf. Harcourt et al., 1981).

Where spiny mice are unusual among muroid rodents is with regard to the spontaneously active luteal phase and long estrous cycles of females (Peitz, 1981). Because stimulation derived from copulation is not critical for initiation of a luteal phase, one might expect reduced complexity of copulation (Bignami and Beach, 1968). Spiny mice seem intermediate in this respect,

however. The number of ejaculations displayed is rather low (see Dewsbury, 1981a) but ejaculation is preceded by a large number of intromissions. It thus appears that male copulatory patterns cannot be predicted from knowledge of the female estrous cycle (for a complementary approach see Dewsbury, 1974b). The multiple intromissions of spiny mice may function in either sperm transport or removal of copulatory plugs (Adler, 1969; Dewsbury, 1981b).

Litters were delivered following 79% of the tests in natural estrus. This suggests that a single ejaculatory series is sufficient for pregnancy in most tests and is consistent with the correlation between male ejaculation frequency and female stimulus requirements proposed by Dewsbury (1978).

Copulatory Behavior of Notomys alexis

Classification of the copulatory pattern of *Notomys alexis* is quite clear-cut, with a true lock, no intravaginal thrusting, ejaculation possible on the first insertion, and copulation usually ceasing at the first ejaculation. This corresponds to pattern 8 of Dewsbury (1972), a pattern shared with pygmy mice, *Batomys taylori* (Estep and Dewsbury, 1976). As with spiny mice, the pattern is little affected by the mode of female estrus. The general nature of male-female interactions and the cessation of copulatory activity with the first ejaculation are consistent with earlier observations (Stanley, 1971; Happold, 1976).

Litters were delivered following 53% of the tests in natural estrus clearly showing that a single lock is sufficient for pregnancy but leaving unanswered the question of conditions favorable for a higher rate. Clearly, a luteal phase can be triggered by a single lock.

To the best of our knowledge, this is the first report of a clear-cut, functional lock in the Muridae. What is particularly remarkable about this is that its occurrence was strongly predicted from various correlations emerging from studies of other muroid species. Locking patterns are correlated with thick glans penes, prominent penile spines, reduced complements of accessory glands, and the lack of a copulatory plug (Dewsbury, 1974a, 1975; Hartung and Dewsbury, 1978). The existence of such characteristics, with the exception of glans thickness, in hopping mice has been repeatedly pointed out by Breed (1980, 1981a, 1981b, 1982; Breed and Sarafis, 1979; Morrissey and Breed, 1982); Morrissey and Breed (1982) addressed the possibility of locking in this species. Such predictive power suggests that the correlations proposed among various characteristics of reproduction in muroid rodents transcend the population or species on which they were formulated and may be somewhat general.

It has further been suggested that a protected nest site is prerequisite to evolution of a locking pattern (Dewsbury, 1975). The extensive burrow systems of hopping mice (Stanley, 1971) are consistent with this expectation.

Among great apes, males in species with low frequencies of copulation tend to have smaller testes than those that copulate more frequently (Short, 1979). Clearly, one must be careful in generalizing from apes to rodents. Nevertheless, the small testes and low ejaculation frequencies of *Notomys alexis* are consistent with this pattern.

Supplementary Data

The supplementary data reported are generally consistent with reports of previous workers and extend those observations in various respects. Morrison et al. (1977) found male spiny mice heavier than females throughout development and Hanney (1965) reported male body length to exceed that of females. Breed (1983) found sexual dimorphism favoring females in hopping mice. The present observations provide quantitative data directly comparable to those from other species (Dewsbury et al., 1980).

The mean age and range of eye opening of 19.6 (18–21) days are consistent with reports of Stanley (1971) that the eyes open between 19 and 21 days and Crichton (1976) that the range is 18–23 days. The precocial nature of spiny mice has often been noted (e.g., Porter and Doane, 1978). Although Strasser (1968) reported that the eyes open on the third day, we found them consistently open on Day 1.

Litter sizes of 2.0 in spiny mice are well within the range reported by other workers (Hanney, 1965; Morrison et al., 1976; Neal, 1983; Strasser, 1968; Young, 1976). The value of 4.4 for hopping mice is somewhat higher than the 2.87 reported by Stanley (1971), but more consistent with the values of 4–5 reported by Watts and Aslin (1981) and Breed (1979). Our observation of a mean gestation period in spiny mice of 38.1 days compares favorably with the 39 day value reported by Strasser (1968), although somewhat higher than the value of a minimum inter-litter interval of 34 days reported by Morrison et al. (1976).

Like Breed (1979, 1981a) we found no copulatory plug in recently-mated hopping mice. However, Breed (1985) reports minute plugs visible in serial sections of the cranial ends of the vaginas of recently-mated females. The presence of a plug in spiny mice is consistent with the observations of coagulation by Peitz et al. (1979). Our observations of the anatomy of the female reproductive tract in *Notomys alexis* are similar to those of Breed (1985).

The Monogamy Scale and Social Organization

The monogamy scale of Dewsbury (1981a) was devised to combine proposed correlates of monogamy that can be measured in the laboratory, especially those discussed by Kleiman (1977), to generate rough predictions regarding the possibility of monogamy in the field. As can be seen in Table 4, there is a considerable difference between the two species under study, with hopping mice achieving a score of +7 and spiny mice of +1. The primary characteristics leading to this difference concern the reversed dimorphism in the two species, the difference in the presence of a plug, and the differential precocity, as reflected in the age at eye opening. The value for hopping mice is quite high, exceeded only by *Peromyscus californicus* among the 24 species for which information on six or more correlates is now available (Dewsbury, 1981a; Sawrey et al., 1984). Nevertheless, in broader perspective, it should be noted that only six of the 23 other species outscore spiny mice on this scale. Thus, application of the scale leads to a strong prediction of monogamy in hopping mice and a weaker prediction in that direction for spiny mice.

There is little information with which to test the validity of this inference for spiny mice. Spiny mice live in both rocky terrain and sandy desserts (Strasser, 1968) but their social patterns in nature have not been well studied. Great care must be exercised in extrapolating from paternal behavior in the laboratory to social structure in the field (Hartung and Dewsbury, 1979). Nevertheless, the extent of male-female interaction and cooperation in the laboratory appears greater than in various other rodents (Dieterlen, 1962; Makin and Porter, 1984) and suggests the possibility of some degree of sociality in the field.

Like male spiny mice, male hopping mice are quite solicitous of both female and young in captivity (Happold, 1976; Stanley, 1971). This sociality is reported to exist in the field with several adults of both sexes sometimes occupying a single burrow system (Breed, 1982; Happold, 1976; Stanley, 1971). It is possible that these may be reproductively active adults socializing during the breeding season, but this is yet to be demonstrated. If these suggestions are confirmed, the sociality of hopping mice predicted from the monogamy scale would be confirmed, but the prediction of monogamy per se would be refuted.

The apparent sociality of *Notomys alexis* is also puzzling with respect to testes sizes. As noted by Harcourt et al. (1981) for primates, small testes tend to characterize species in which one male mates with each female. Large testes may be seen as an adaptation to sperm competition in multi-male breeding systems (see Dewsbury, 1984). However, according to Breed (1982:612) "A promiscuous reproductive strategy seems likely for this species." Further detailed study of hopping mice in the field will be needed to provide a definitive test of the prediction of monogamy.

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